

PROJECT PROPOSAL:

ReliefLink: Disaster Relief Distribution and Priority Management System

Course/Section: BSIT 2B

Project Category: Web-Based Integrated Disaster Relief Management System

Date: April 7, 2026

PROJECT TEAM

Lead Integration Architect: Juan, Prince Jheck T.– *Responsible for API connectivity and system logic.*

Backend Developer: Ramores, Jamaica– *Responsible for Database and Server-side scripting.*

Frontend Developer: Rebanal, Jordan Mike – *Responsible for UI/UX and client-side validation.*

PROJECT DESCRIPTION

Current Situation:

Currently, disaster response in many municipalities relies on manual reporting and coordination, where barangay officials submit information through paper-based processes or messaging platforms. This approach often leads to delays in data collection, inconsistencies in reported information, and difficulty in verifying the accuracy of reports. As a result, the distribution of relief goods becomes inefficient, with some highly affected barangays receiving assistance later than needed, while others may receive disproportionate aid due to lack of proper prioritization.

The Proposed System:

:

The proposed system, **ReliefLink**, addresses the inefficiencies of manual disaster response by automating the collection, processing, and analysis of disaster-related data. Barangay officials can submit reports through a centralized web platform, eliminating delays and reducing errors associated with manual reporting. The system automatically processes the submitted data and prioritizes barangays based on the severity of impact and number of affected residents, ensuring that relief goods are distributed efficiently and fairly.

From an integrative perspective, the system connects the internal municipal database with an external **weather API**, allowing real-time disaster information to be combined with local reports. Through this integration, the PHP-based middleware synchronizes and processes data from both sources using JSON, enabling more accurate decision-making. This automated and integrated approach ensures faster response times, improved data accuracy, and more effective coordination in disaster relief operations.

SDG Alignment:

This project supports **SDG 11: Sustainable Cities and Communities** by improving disaster response efficiency, ensuring timely delivery of relief goods, and enhancing community resilience through technology.

TECHNICAL ARCHITECTURE

This section defines the "Integrative" nature of the project.

Table 1. Title

Component	Specification
Platform	Web-based Application (Responsive)
Frontend Stack	HTML5, CSS3 JavaScript (ES6+)
Backend Stack	PHP 8.x
Database Management	MySQL (Relational Database)
Primary API Integration	OpenWeather API
Data Exchange Format	JSON / RESTful Request
Integration Function	Fetch weather data and combine with barangay reports to prioritize relief distribution

Discussion:

The system uses **PHP as a middleware** to connect the internal database with the external **OpenWeather API**. Barangay disaster reports are stored in MySQL, while real-time weather data is fetched through API requests and processed using JSON format. By integrating these two data sources, the system can automatically analyze disaster conditions and identify the most affected barangays. This allows the municipality to make faster, data-driven decisions and ensures a more efficient and prioritized distribution of relief goods.

FUNCTIONAL REQUIREMENTS

A. User Module (Barangay Officials)

The system shall allow barangay officials to register and securely log in to the platform. Users can submit disaster reports including the number of affected families, damaged houses, and disaster type. They can also view submitted reports and receive updates or notifications regarding relief distribution and report status.

B. Administrative Module (MDRRMO)

The system shall provide administrators with a dashboard for monitoring disaster reports and affected barangays. Admin users can manage relief inventory, including adding, updating, and tracking available supplies. The system shall automatically prioritize barangays based on submitted data and display results for decision-making. Administrators can also monitor API data, view system logs, and oversee the distribution of relief goods.

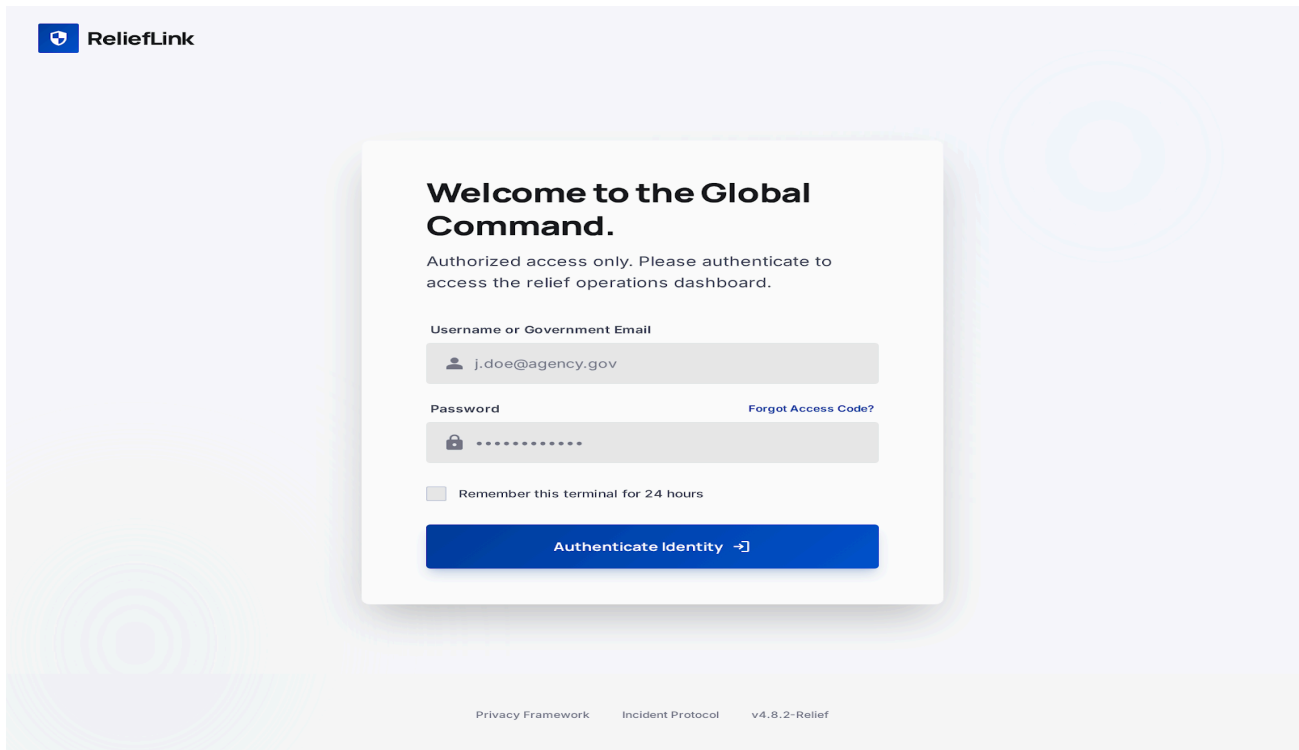
UI/UX PROTOTYPE & DESIGN

Since you are including a UI prototype, it is helpful to visualize how the integration works behind the scenes.

Key Wireframes (Screenshots)

Place your mockup images here. For IPT projects, you must include the following four critical screens:

PUBLIC / AUTHENTICATION INTERFACE



The image shows the 'Welcome to the Global Command.' login page for ReliefLink. It features a central white card on a light blue background with faint circular patterns. The card contains the ReliefLink logo, a welcome message, and a login form. The form includes fields for 'Username or Government Email' (with the example 'j.doe@agency.gov') and 'Password' (masked with dots). A 'Forgot Access Code?' link is next to the password field. Below the password field is a checkbox for 'Remember this terminal for 24 hours'. At the bottom of the card is a blue button labeled 'Authenticate Identity' with a right-pointing arrow. At the very bottom of the page, there are three links: 'Privacy Framework', 'Incident Protocol', and 'v4.8.2-Relief'.

ReliefLink

Welcome to the Global Command.

Authorized access only. Please authenticate to access the relief operations dashboard.

Username or Government Email

j.doe@agency.gov

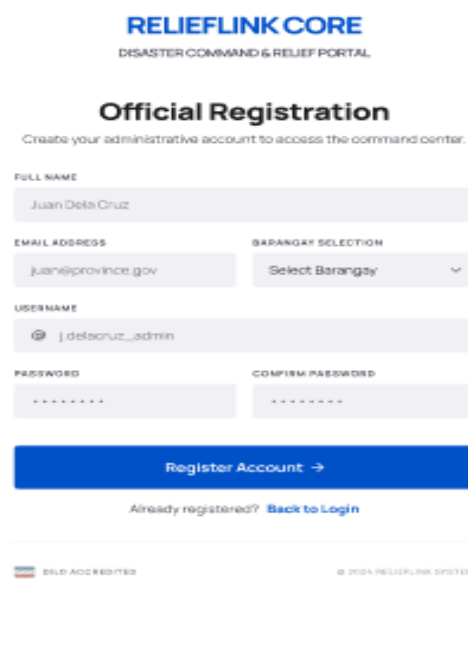
Password [Forgot Access Code?](#)

☐ Remember this terminal for 24 hours

Authenticate Identity →

[Privacy Framework](#) [Incident Protocol](#) [v4.8.2-Relief](#)

Figure 1. Login & Authentication



The image shows the 'Official Registration' page for 'RELIEFLINK CORE', the 'DISASTER COMMAND & RELIEF PORTAL'. The page has a white background with a blue header. It includes a title 'Official Registration' and a subtitle 'Create your administrative account to access the command center.' The registration form consists of several fields: 'FULL NAME' (filled with 'Juan Dela Cruz'), 'EMAIL ADDRESS' (filled with 'juan@province.gov'), 'BARANGAY SELECTION' (a dropdown menu showing 'Select Barangay'), 'USERNAME' (filled with 'jdelacruz_admin'), 'PASSWORD' (masked with dots), and 'CONFIRM PASSWORD' (masked with dots). Below the form is a blue button labeled 'Register Account' with a right-pointing arrow. Underneath the button is a link: 'Already registered? [Back to Login](#)'. At the bottom of the page, there is a small 'BID-AACREDITED' logo and the text '© 2024 RELIEFLINK SYSTEM'. In the bottom right corner, there is a small blue icon and the text 'SECURITY CENTER'.

RELIEFLINK CORE
DISASTER COMMAND & RELIEF PORTAL

Official Registration

Create your administrative account to access the command center.

FULL NAME

Juan Dela Cruz

EMAIL ADDRESS

juan@province.gov

BARANGAY SELECTION

Select Barangay

USERNAME

jdelacruz_admin

PASSWORD

CONFIRM PASSWORD

Register Account →

Already registered? [Back to Login](#)

 BID-AACREDITED

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 SECURITY CENTER

Figure 2. Registration Page (Barangay Officials only)

BARANGAY OFFICIAL INTERFACE (User Side)

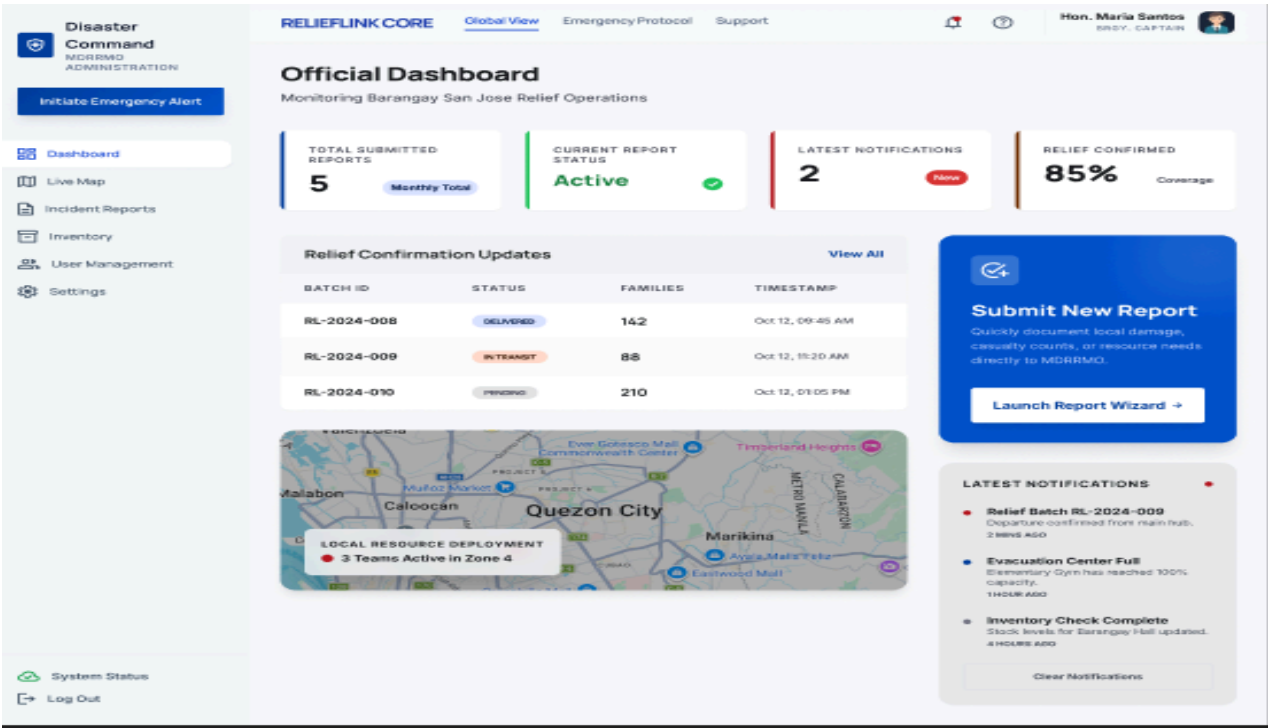


Figure 3. User Dashboard Page

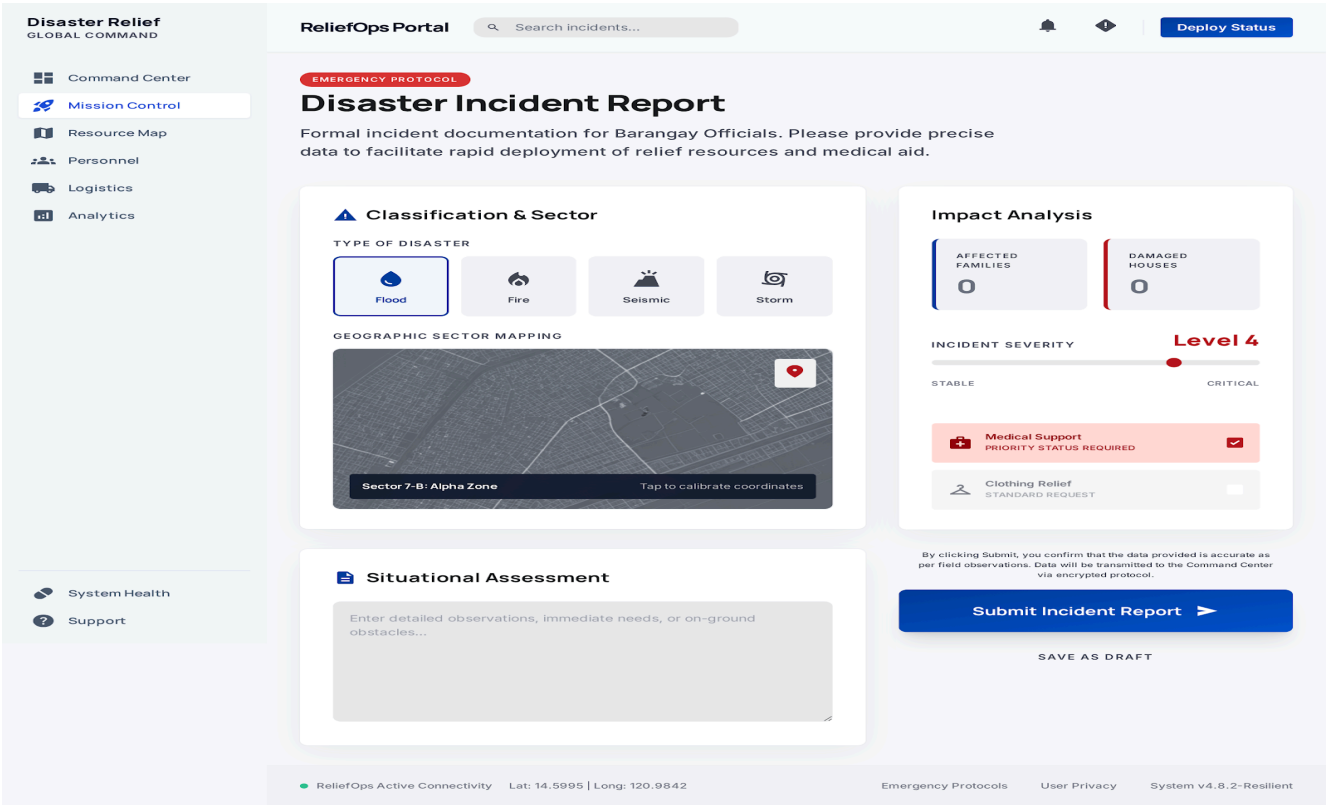


Figure 4. Submit Disaster Report Page

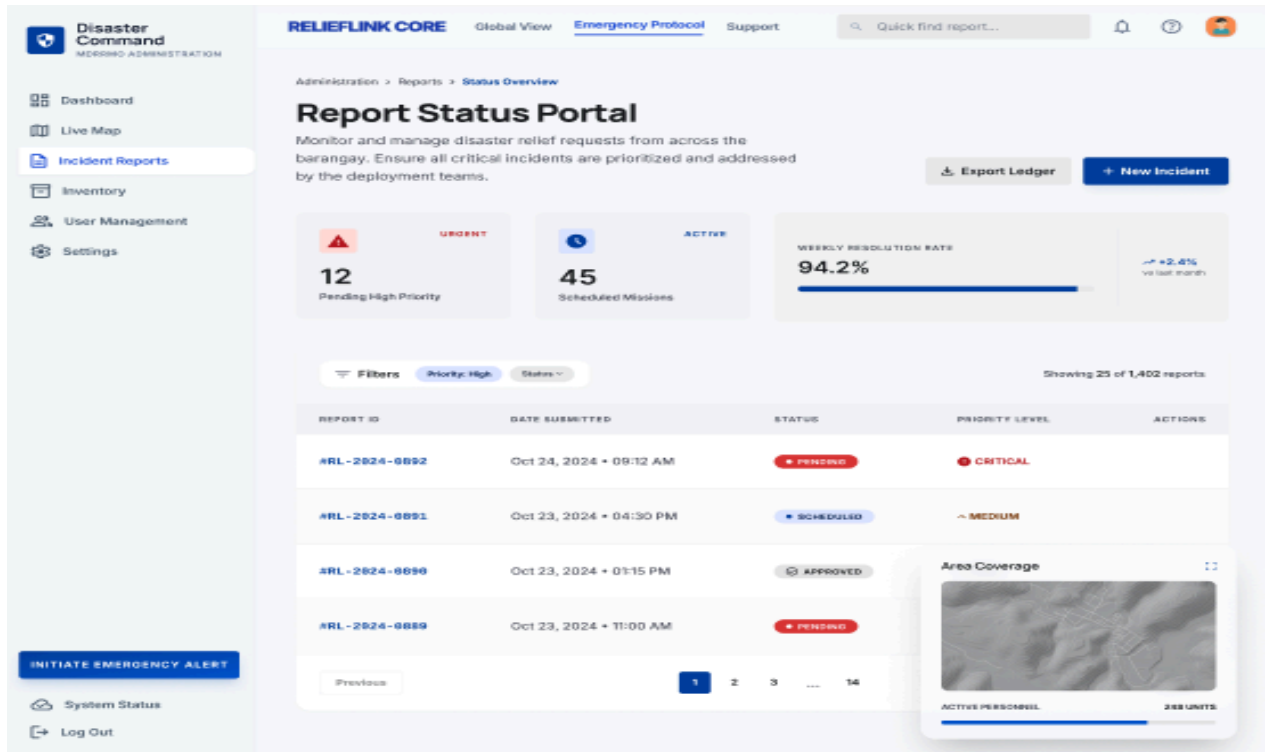


Figure 5. View Report Status Page

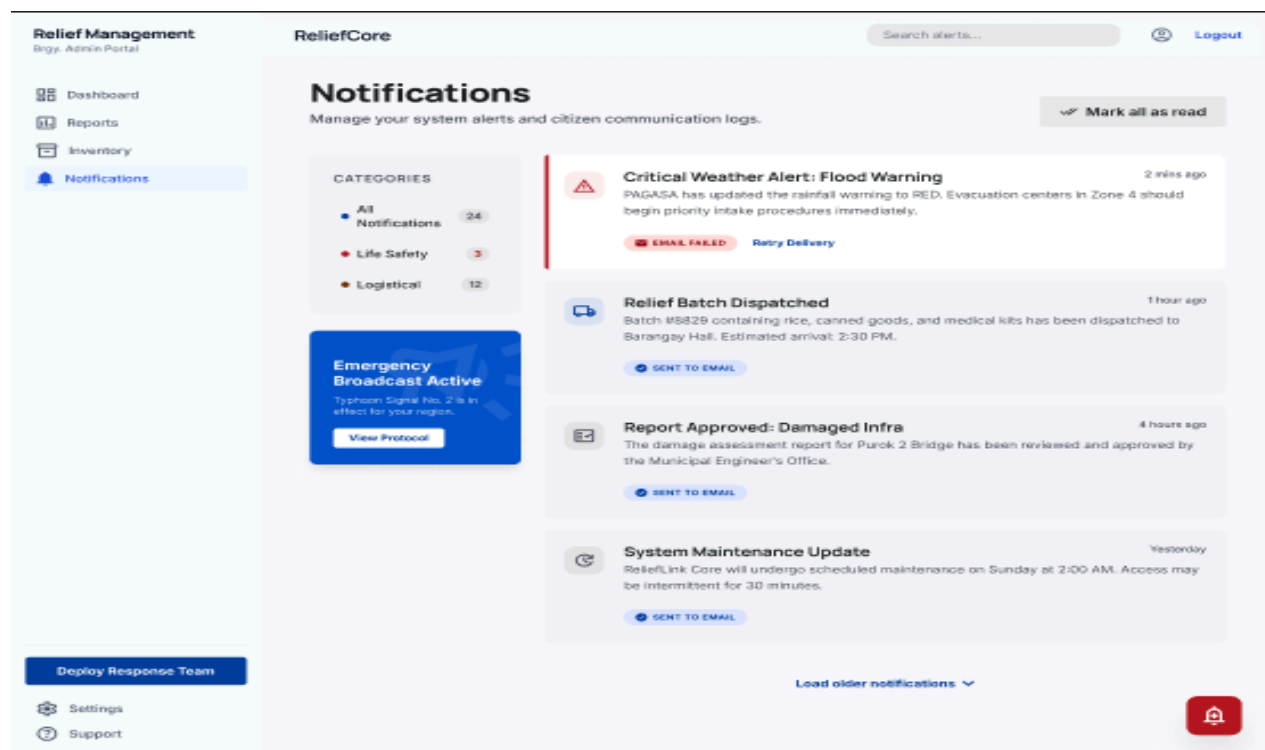


Figure 6. Notifications Page

ADMIN (MDRRMO) INTERFACE

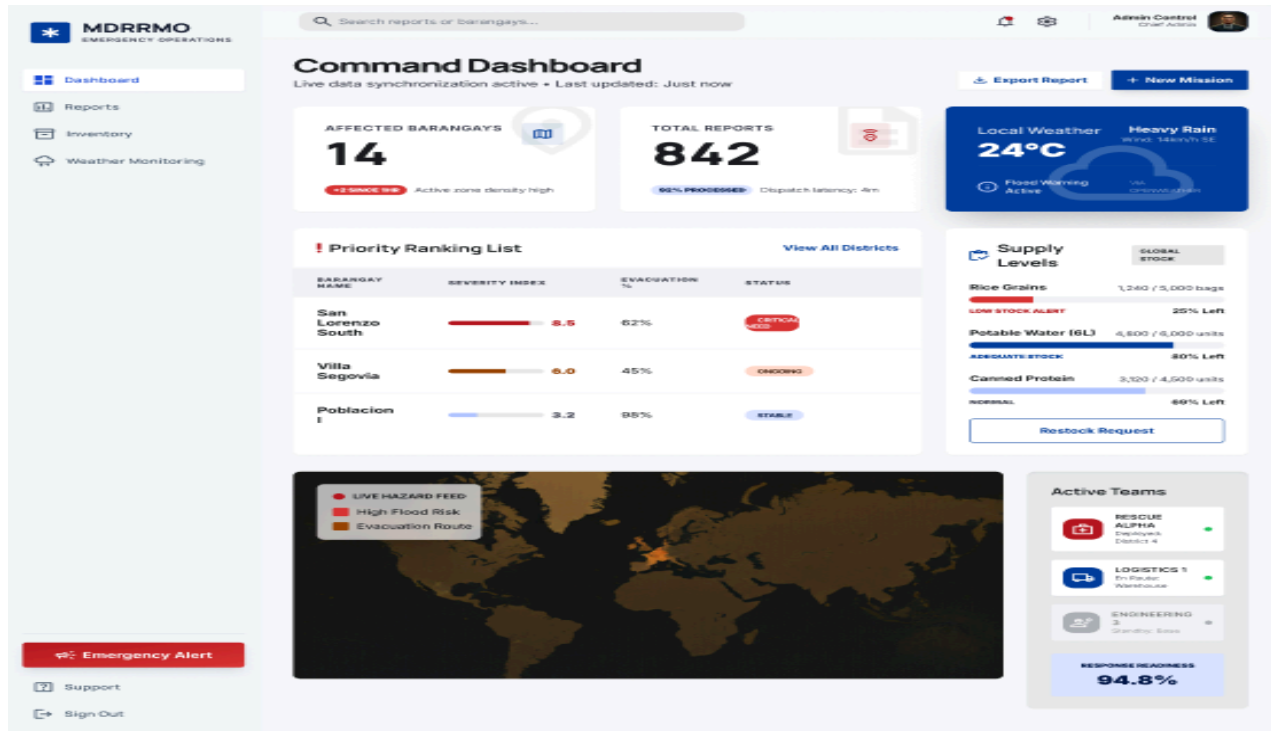


Figure 7. Admin Dashboard

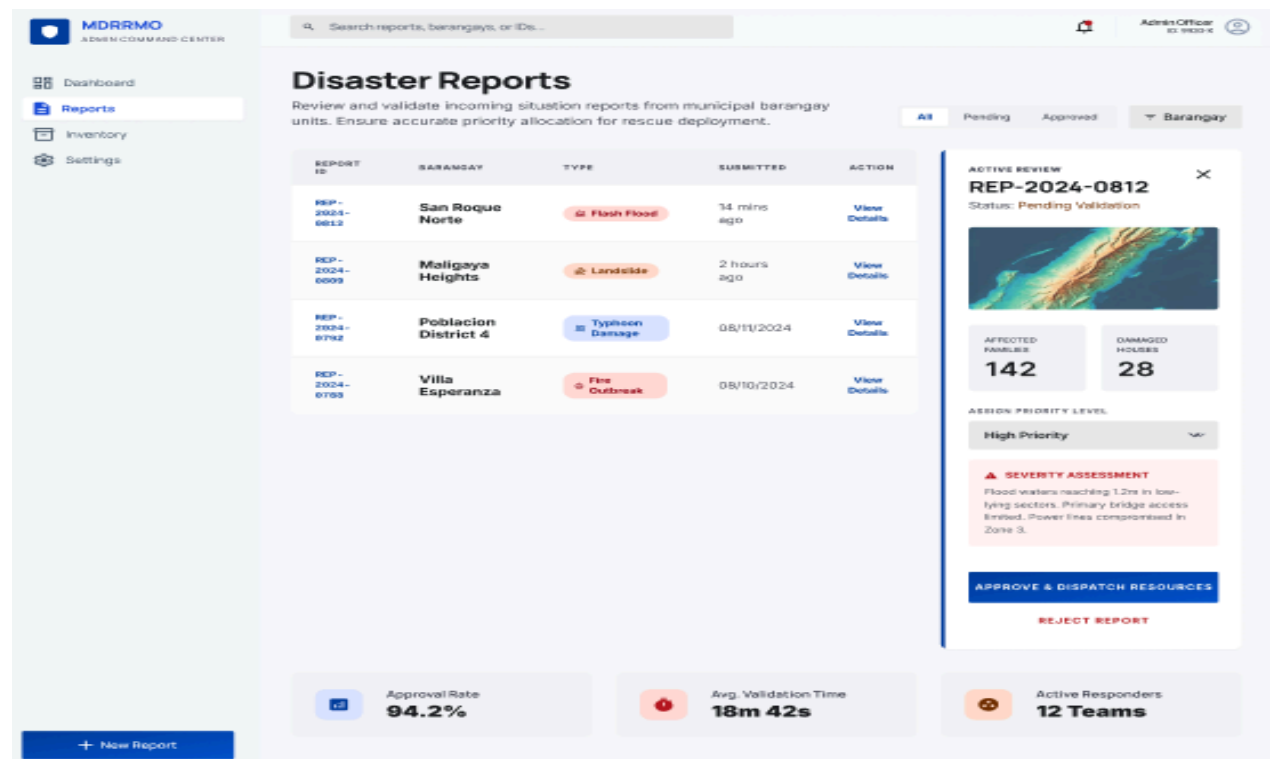


Figure 8. Review Disaster Reports Page

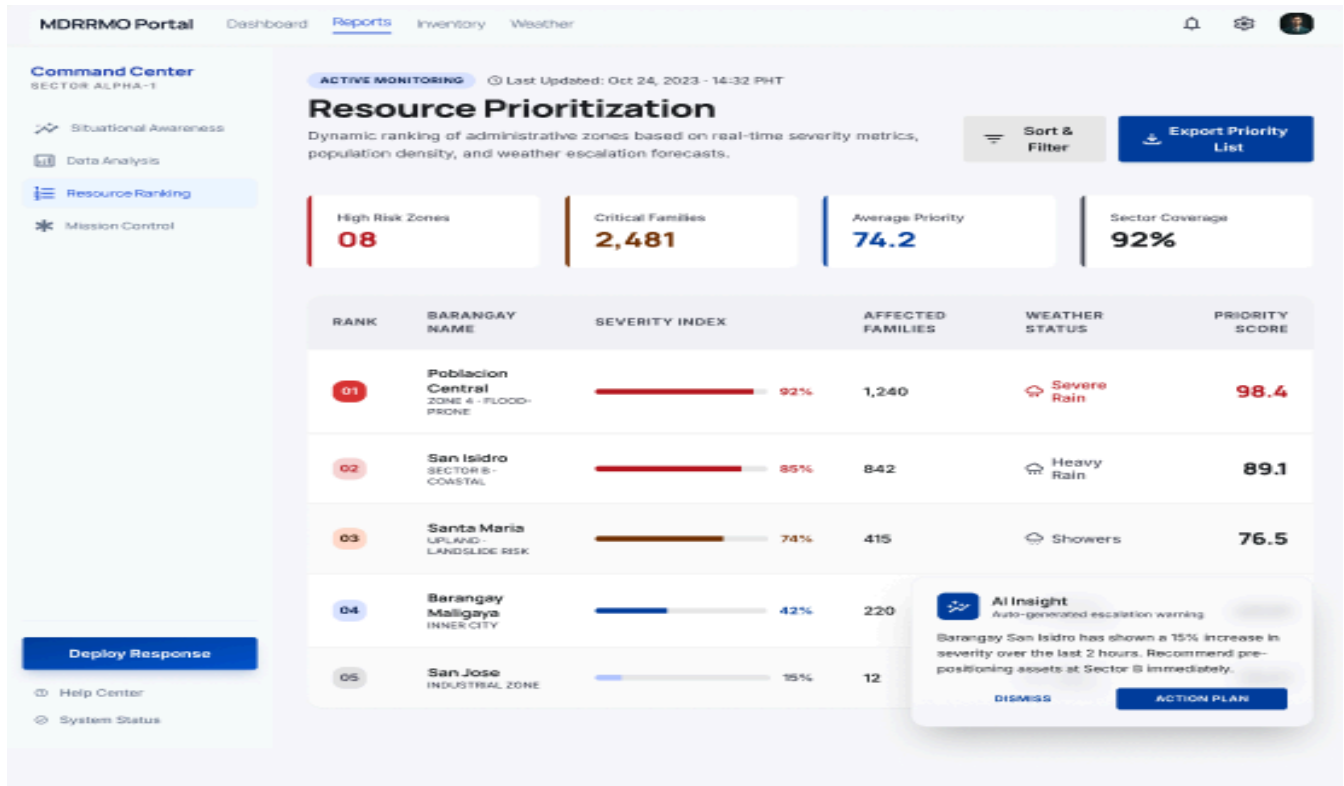


Figure 9. Prioritize Barangays Page

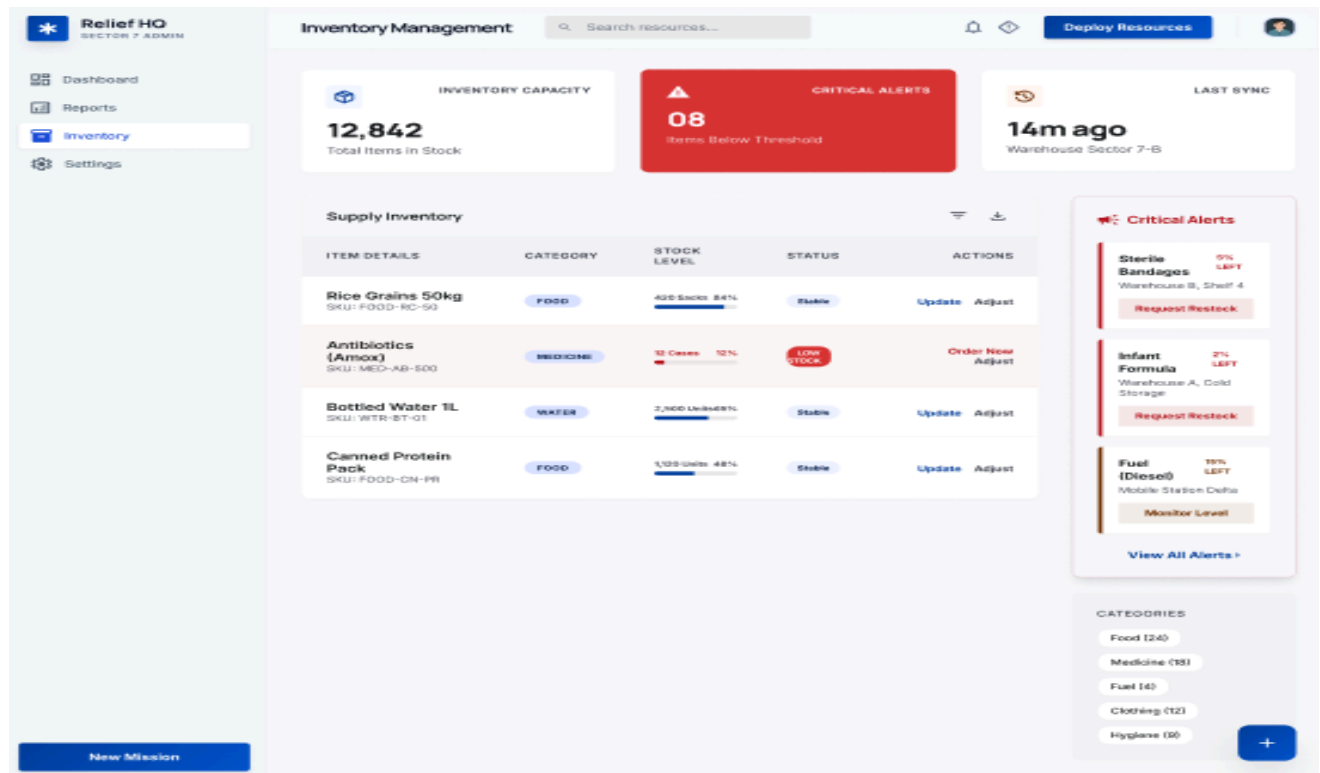


Figure 10. Relief Inventory Management Page

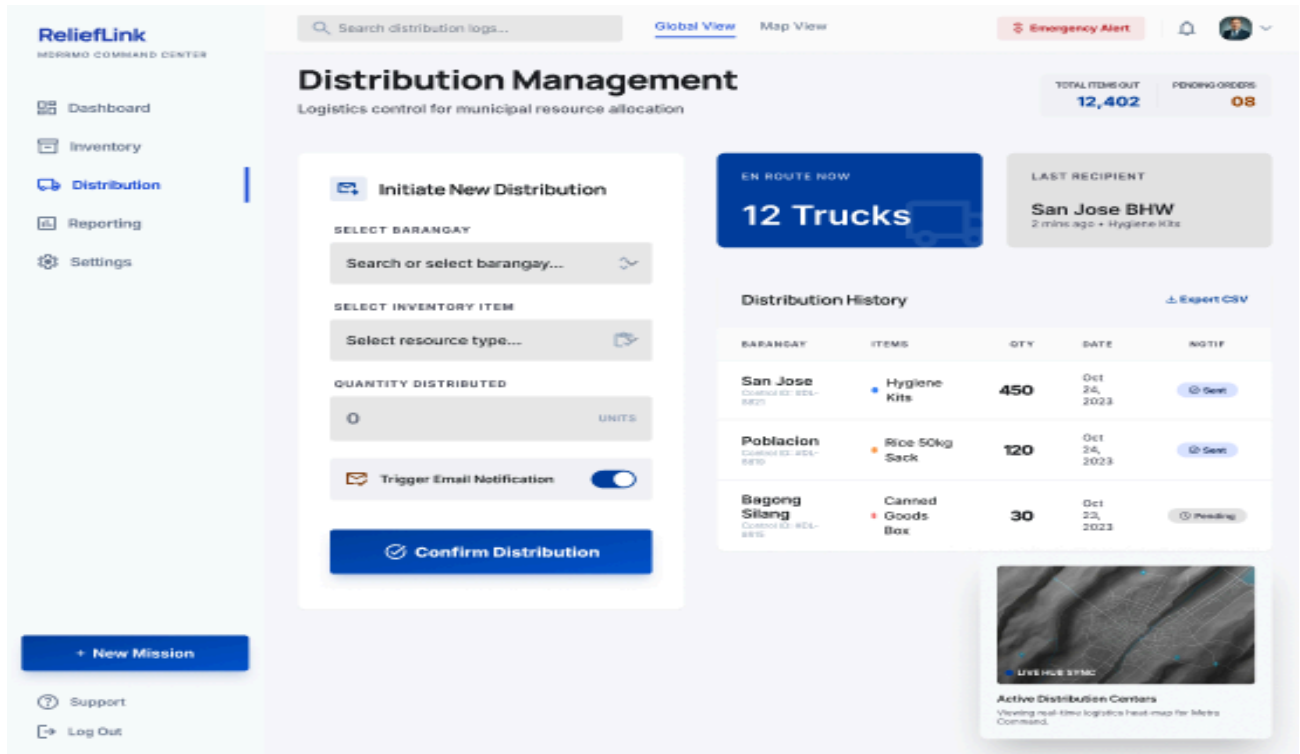


Figure 11. Distribution Management Page

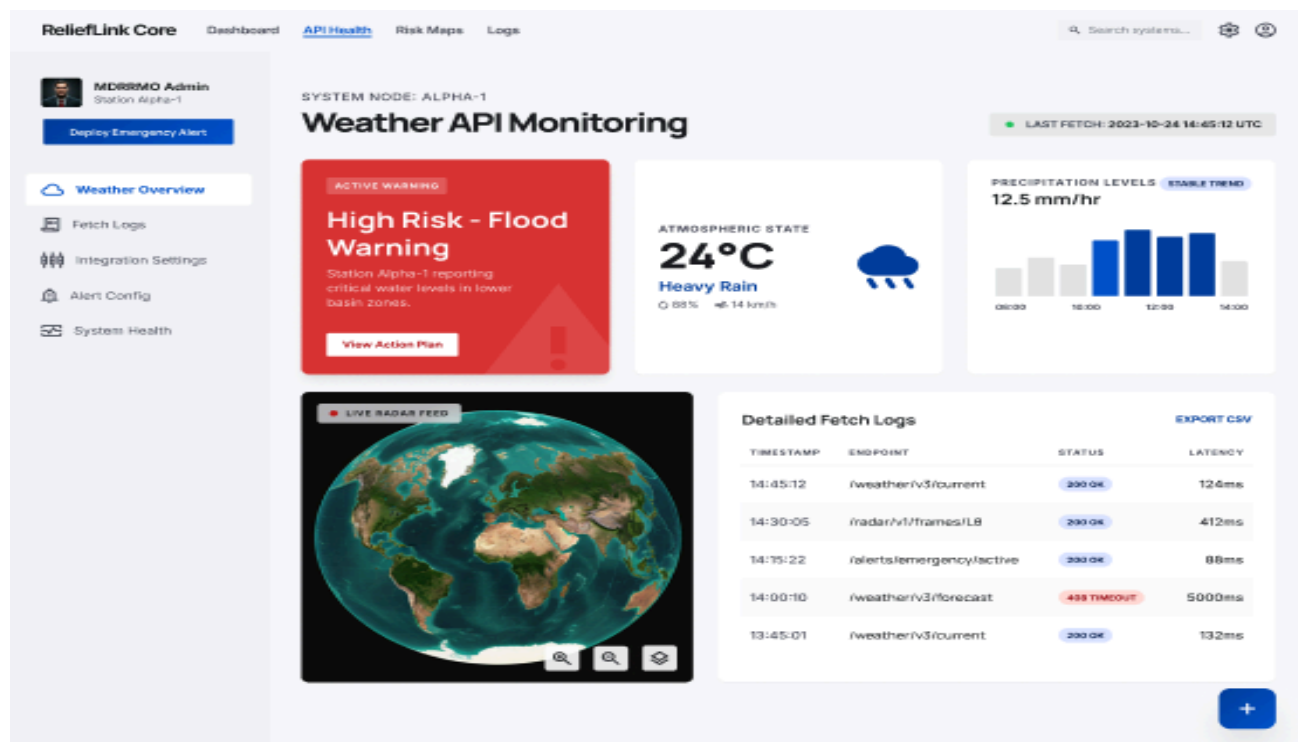


Figure 12. Weather API Monitoring Page

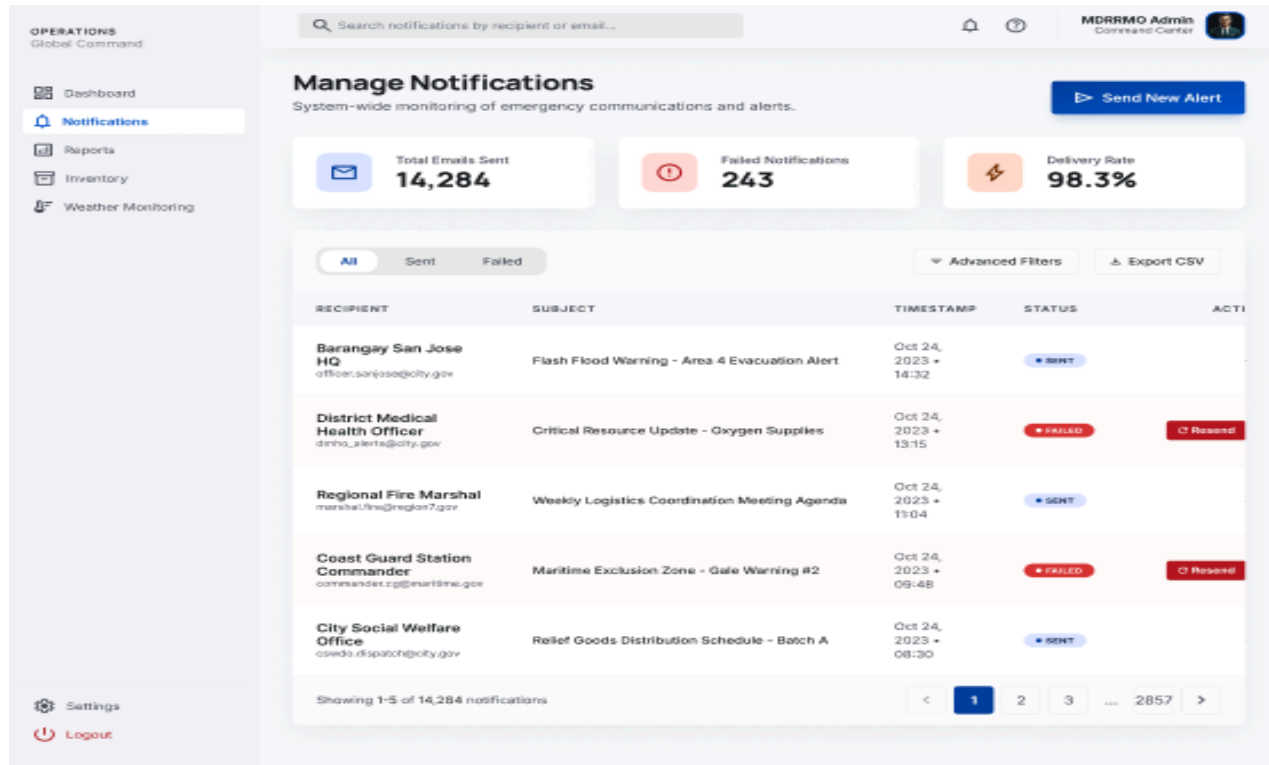


Figure 13. Manage Notifications Page

User Experience (UX) Flowchart

A diagram showing how a user moves from the Landing Page -> Login -> Performing a specific Task (like requesting a document) -> Receiving a Notification.

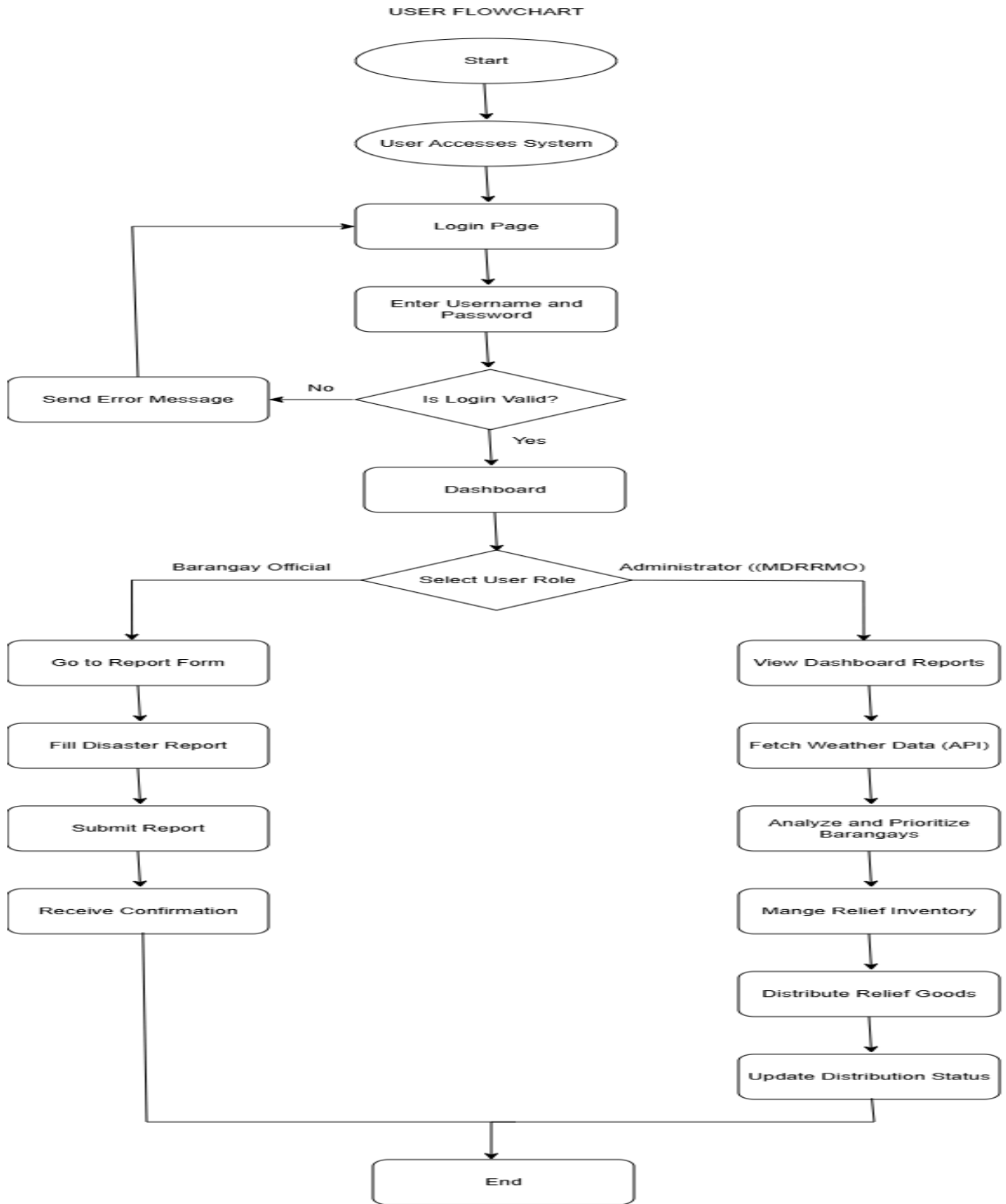


Figure 3. Flowchart

SYSTEM DESIGN (DIAGRAMS)

Entity Relationship Diagram (ERD)

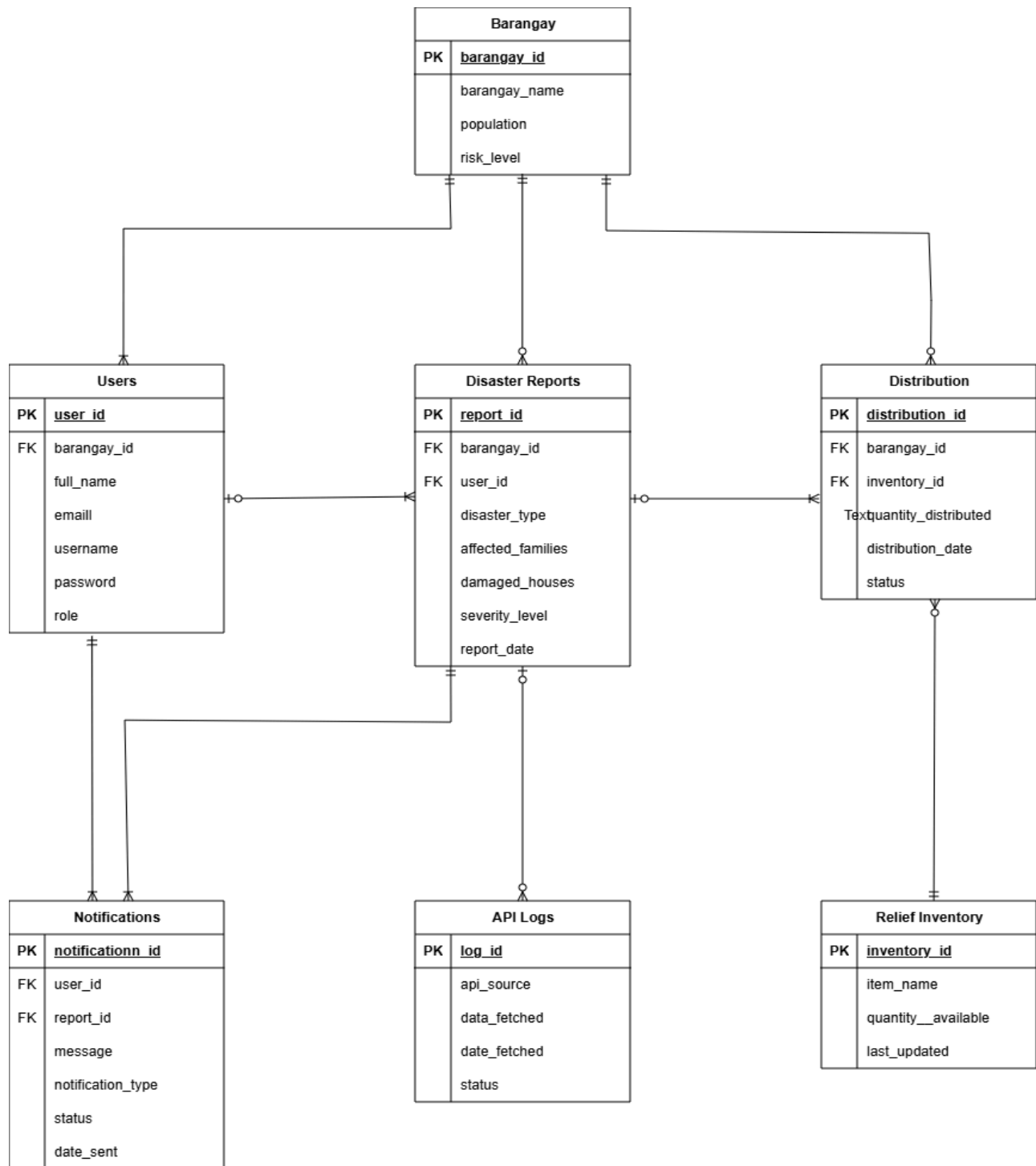


Figure 4. Entity Relationship Diagram

Figure 5. Use Case Diagram of ReliefLink System

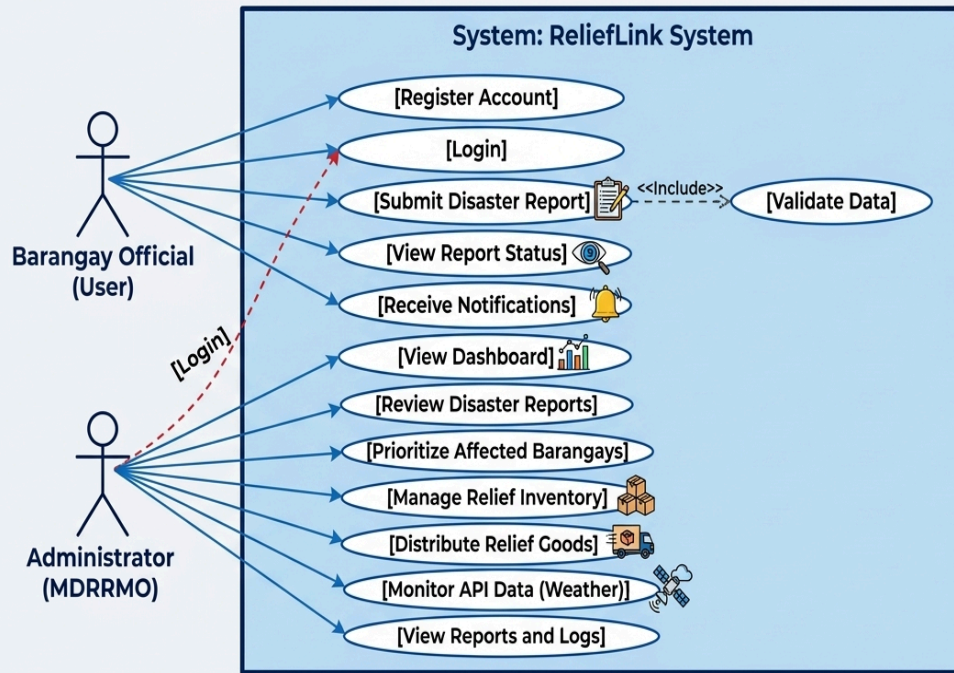


Figure 5. Use Case Diagram Of ReliefLink System

VI. DEVELOPMENT TIMELINE

Phase	Activities	Duration
Planning	Requirements gathering and ERD Design	1 Week
Development	Frontend design, Backend coding, Database setup	4 Weeks
Integration	API integration and Middleware testing	2 Weeks
Deployment	User testing and Final Demo preparation	1 Week

Discussion:

The project will follow a structured development process starting with planning and system design. The development phase focuses on building the frontend and backend components, followed by integration of the external API using PHP middleware. Testing ensures that all modules function correctly before final deployment and presentation.

Instructions for filling this out:

PRESENTATION REQUIREMENT (PPT)

IMPORTANT: All teams are required to create a PowerPoint Presentation for the proposal defense. The PPT must include:

1. Clear identification of the Lead Architect, Backend, and Frontend developers.
2. A demonstration of the UI flow using a tool like Figma or a sequence of screenshots.
3. A slide specifically explaining the PHP code or logic that handles the Primary API Integration.
4. Ensure the "Target Users" and "Features" match the entries previously submitted in the IPT Excel Project Registry.

Instructions for Submission:

1. Make sure your tables include primary keys (ID) and foreign keys to show how data links together.
2. Be very specific about which API you are using and how your PHP code handles the response.
3. Match the target user with what you have in your Excel file to ensure consistency.
4. Submit this proposal in PDF format and the accompanying PPT file before the scheduled defense.